2D Explicit FEA — Copper Panel Impact

MATLAB source listing: copper_panel_impact_fea.m
441 lines | plane-stress Q4 elements, explicit time integration

Impact 1: steel debris (r = 2.5 mm, 1000 m/s) onto a C10100 copper panel
Impact 2: steel debris onto the top wall of a Ti-6Al-4V frame

Requires MATLAB R2020b or newer (uses turbo, yline, gobjects).
See README.md and THEORY.md in the repository for usage and the underlying theory.

copper_panel_impact_fea.m

```
%% ============================================================
%  2D Explicit FEA — COPPER PANEL IMPACT
%  ============================================================
%  Impact 1 : Steel debris → COPPER right panel
%  Impact 2 : Steel debris → Titanium Ti-6Al-4V frame top wall
% ============================================================
clear; close all; clc;

fprintf('================================================\n');
fprintf('  DUAL IMPACT FEA — COPPER PANEL\n');
fprintf('================================================\n');

%% ===== MATERIALS =====
% COPPER C10100
E_cu  = 115e9;   nu_cu = 0.35;  rho_cu = 8960;
cw_cu = sqrt(E_cu/rho_cu);

% Titanium Ti-6Al-4V
E_ti  = 116e9;   nu_ti = 0.34;  rho_ti = 4430;
cw_ti = sqrt(E_ti/rho_ti);

% Steel debris
rho_d = 7850;
R_d   = 2.5e-3;
m_d   = (4/3)*pi*R_d^3*rho_d;

%% ===== GEOMETRY =====
frame_outer = 200;    % mm
frame_wall  = 10;     % mm
frame_inner = frame_outer - 2*frame_wall;
panel_L     = 280;    % mm
panel_H     = 20;     % mm
panel_th_cu = 1.32e-3; % m
frame_th_ti = 10e-3;   % m

%% ===== D-MATRICES =====
Dm_cu = E_cu/(1-nu_cu^2)*[1 nu_cu 0; nu_cu 1 0; 0 0 (1-nu_cu)/2];
Dm_ti = E_ti/(1-nu_ti^2)*[1 nu_ti 0; nu_ti 1 0; 0 0 (1-nu_ti)/2];

%% ===== MESH 1 — COPPER panel =====
NX_p = 80;  NY_p = 10;
Lx_p = panel_L * 1e-3;
Ly_p = panel_H * 1e-3;
ex_p = Lx_p/NX_p;  ey_p = Ly_p/NY_p;

px0 = (frame_outer/2)*1e-3;
py0 = -(panel_H/2)*1e-3;

[XI_p,YI_p] = meshgrid(linspace(0,Lx_p,NX_p+1), linspace(0,Ly_p,NY_p+1));
Xp = px0 + XI_p(:);
Yp = py0 + YI_p(:);
nn_p = numel(Xp);

elem_p = zeros(NX_p*NY_p, 4);
idx=0;
for j=1:NY_p
  for i=1:NX_p
    idx=idx+1;
    n1=(j-1)*(NX_p+1)+i;
    elem_p(idx,:)=[n1, n1+1, n1+(NX_p+1)+1, n1+(NX_p+1)];
  end
end
ne_p = size(elem_p,1);

Bp = zeros(3,8);
dNdx_p=(1/(2*ex_p))*[-1 1 1 -1]; dNdy_p=(1/(2*ey_p))*[-1 -1 1 1];
Bp(1,1:2:end)=dNdx_p; Bp(2,2:2:end)=dNdy_p;
Bp(3,1:2:end)=dNdy_p; Bp(3,2:2:end)=dNdx_p;
Ve_p = ex_p*ey_p*panel_th_cu;

rho2d_cu = rho_cu*panel_th_cu;
Mp = zeros(nn_p,1);
for e=1:ne_p
  en=elem_p(e,:);
  Mp(en)=Mp(en)+rho2d_cu*(ex_p*ey_p)/4;
end
bc_p = find(abs(Xp-px0)<1e-9);
Mp(bc_p) = Mp(bc_p)*200;
```

```
79
80 k_pen_cu = E_cu * R_d * 32;
81
82 %% ===== MESH 2 — Titanium frame =====
83 fw = frame_outer/2;   fh = frame_outer/2;
84 NX_f = 50;  NY_f = 5;
85 Lx_f = frame_outer*1e-3;
86 Ly_f = frame_wall*1e-3;
87 ex_f = Lx_f/NX_f;  ey_f = Ly_f/NY_f;
88
89 fx0 = -fw*1e-3;
90 fy0 = (fh-frame_wall)*1e-3;
91
92 [XI_f,YI_f] = meshgrid(linspace(0,Lx_f,NX_f+1), linspace(0,Ly_f,NY_f+1));
93 Xf = fx0 + XI_f(:);
94 Yf = fy0 + YI_f(:);
95 nn_f = numel(Xf);
96
97 elem_f = zeros(NX_f*NY_f, 4);
98 idx=0;
99 for j=1:NY_f
100   for i=1:NX_f
101     idx=idx+1;
102     n1=(j-1)*(NX_f+1)+i;
103     elem_f(idx,:)=[n1, n1+1, n1+(NX_f+1)+1, n1+(NX_f+1)];
104   end
105 end
106 ne_f = size(elem_f,1);
107
108 Bf = zeros(3,8);
109 dNdx_f=(1/(2*ex_f))*[-1 1 1 -1]; dNdy_f=(1/(2*ey_f))*[-1 -1 1 1];
110 Bf(1,1:2:end)=dNdx_f; Bf(2,2:2:end)=dNdy_f;
111 Bf(3,1:2:end)=dNdy_f; Bf(3,2:2:end)=dNdx_f;
112 Ve_f = ex_f*ey_f*frame_th_ti;
113
114 rho2d_f = rho_ti*frame_th_ti;
115 Mf = zeros(nn_f,1);
116 for e=1:ne_f
117   en=elem_f(e,:);
118   Mf(en)=Mf(en)+rho2d_f*(ex_f*ey_f)/4;
119 end
120 bc_f = find( (abs(Xf-fx0)<1e-9) | (abs(Xf-(fx0+Lx_f))<1e-9) );
121 Mf(bc_f) = Mf(bc_f)*200;
122
123 k_pen_f = E_ti * R_d * 32;
124
125 %% ===== TIMESTEP =====
126 Lmin = min([ex_p, ey_p, ex_f, ey_f]);
127 dt   = 0.80 * Lmin / max(cw_cu, cw_ti);
128 T_end  = 650e-6;
129 Nsteps = ceil(T_end/dt);
130
131 fprintf('dt       = %.3f ns\n', dt*1e9);
132 fprintf('T_end    = %.0f µs\n', T_end*1e6);
133 fprintf('Steps    = %d\n\n', Nsteps);
134
135 %% ===== STATE VECTORS =====
136 up  = zeros(nn_p,2);  vp = zeros(nn_p,2);
137 uf  = zeros(nn_f,2);  vf = zeros(nn_f,2);
138 svm_p = zeros(ne_p,1);
139 svm_f = zeros(ne_f,1);
140
141 %% ===== DEBRIS INITIAL CONDITIONS =====
142 dX1 = px0 + Lx_p/2;
143 dY1 = py0 + Ly_p + R_d + 0.002;
144 dVx1=0;  dVy1=-1000;
145 wave_r1=0; imp_x1=dX1; imp_y1=py0+Ly_p;
146 done1=false; contact1=false;
147
148 dX2 = fx0 + Lx_f/2;
149 dY2 = fy0 + Ly_f + R_d + 0.002;
150 dVx2=0;  dVy2=-1000;
151 wave_r2=0; imp_x2=dX2; imp_y2=fy0+Ly_f;
152 done2=false; contact2=false;
153
154 %% ===== HISTORY ARRAYS =====
155 def_hist_p = zeros(Nsteps,1);
156 def_hist_f = zeros(Nsteps,1);
```

```
t_hist     = zeros(Nsteps,1);
max_def_p  = 0;
max_def_f  = 0;

%% ===== FIGURE SETUP =====
fig = figure('Color',[0.03 0.05 0.09], ...
             'Position',[20 20 1500 850], ...
             'Name','Dual Impact FEA - Copper Panel', ...
             'NumberTitle','off');

%% Main scene
ax = axes('Parent',fig,'Position',[0.03 0.18 0.58 0.78]);
set(ax,'Color',[0.03 0.05 0.09],'XColor',[0.55 0.70 0.88], ...
       'YColor',[0.55 0.70 0.88],'GridColor',[0.12 0.20 0.30], ...
       'GridAlpha',0.35,'Box','on');
hold(ax,'on'); axis(ax,'equal'); grid(ax,'on');
xlabel(ax,'x  (mm)','Color',[0.55 0.70 0.88],'FontSize',10);
ylabel(ax,'y  (mm)','Color',[0.55 0.70 0.88],'FontSize',10);
title(ax,' DUAL IMPACT FEA  |  COPPER Panel + Ti Frame  |  Von Mises stress', ...
      'Color',[0.88 0.94 1.00],'FontSize',11,'FontWeight','bold');
xlim(ax,[-440 610]); ylim(ax,[-180 280]);

%% Deformation plots
ax2 = axes('Parent',fig,'Position',[0.65 0.57 0.16 0.36]);
set(ax2,'Color',[0.03 0.05 0.09],'XColor',[0.55 0.70 0.88], ...
        'YColor',[0.55 0.70 0.88]);
hold(ax2,'on'); grid(ax2,'on');
xlim(ax2,[0 650]); ylim(ax2,[0 45]);
xlabel(ax2,'Time  (µs)','Color',[0.55 0.70 0.88],'FontSize',8);
ylabel(ax2,'Max deflection (mm)','Color',[0.55 0.70 0.88],'FontSize',8);
title(ax2,'COPPER Panel','Color',[0.85 0.65 0.25],'FontSize',9);

ax4 = axes('Parent',fig,'Position',[0.83 0.57 0.16 0.36]);
set(ax4,'Color',[0.03 0.05 0.09],'XColor',[0.55 0.70 0.88], ...
        'YColor',[0.55 0.70 0.88]);
hold(ax4,'on'); grid(ax4,'on');
xlim(ax4,[0 650]); ylim(ax4,[0 3]);
xlabel(ax4,'Time  (µs)','Color',[0.55 0.70 0.88],'FontSize',8);
ylabel(ax4,'Max deflection (mm)','Color',[0.55 0.70 0.88],'FontSize',8);
title(ax4,'Titanium Frame','Color',[0.60 0.95 0.70],'FontSize',9);

ax3 = axes('Parent',fig,'Position',[0.65 0.10 0.34 0.38]);
set(ax3,'Color',[0.03 0.05 0.09],'XColor',[0.55 0.70 0.88], ...
        'YColor',[0.55 0.70 0.88]);
hold(ax3,'on'); grid(ax3,'on');
xlim(ax3,[0 650]); ylim(ax3,[0 30]);
xlabel(ax3,'Time  (µs)','Color',[0.55 0.70 0.88],'FontSize',8);
ylabel(ax3,'Error (%)','Color',[0.55 0.70 0.88],'FontSize',8);
title(ax3,'Calibration Error','Color',[0.88 0.94 1.0],'FontSize',9);
yline(ax3,15,'--r','LineWidth',1.2,'Label','15% limit');
yline(ax3,0,'--w','LineWidth',0.8);

colormap(ax, turbo(512));

%% Draw static structure
ti_col  = [0.10 0.22 0.16];
ti_edge = [0.32 0.72 0.50];
cu_col  = [0.40 0.20 0.10];
cu_edge = [0.85 0.55 0.25];

% Titanium frame
fill(ax,[-fw fw fw -fw -fw],[-fh -fh fh fh -fh], ...
     ti_col,'EdgeColor',ti_edge,'LineWidth',1.8);
fi = frame_inner/2;
fill(ax,[-fi fi fi -fi -fi],[-fi -fi fi fi -fi], ...
     [0.03 0.05 0.09],'EdgeColor',[0.20 0.50 0.35]);
line(ax,[-fw fw],[0 0],'Color',ti_edge);
line(ax,[0 0],[-fh fh],'Color',ti_edge);
text(ax,0,-6,'Titanium frame','Color',[0.55 0.92 0.68],'FontSize',8,...
     'HorizontalAlignment','center');

% Left panel
fill(ax,[-fw-panel_L -fw -fw -fw-panel_L],...
        [-panel_H/2 -panel_H/2 panel_H/2 panel_H/2],...
     [0.07 0.18 0.38],'EdgeColor',[0.38 0.62 0.90],'FaceAlpha',0.6);
text(ax,-fw-panel_L/2, panel_H/2+12,'Left panel (static)',...
     'Color',[0.48 0.70 0.95],'FontSize',7,'HorizontalAlignment','center');

```

```
% Connecting rods
line(ax,[-fw-panel_L -fw],[0 0],'Color',[0.38 0.62 0.90],'LineWidth',2.5);
line(ax,[fw fw+panel_L],[0 0],'Color',cu_edge,'LineWidth',2.5);

% Right Copper panel label
text(ax, fw+panel_L/2, panel_H/2+12,'Right panel (COPPER)',...
     'Color',[0.85 0.65 0.35],'FontSize',7,'HorizontalAlignment','center');
text(ax, fw+panel_L/2, panel_H/2+22,'C10100 | E=115 GPa, ρ=8960 kg/m³',...
     'Color',[0.85 0.55 0.25],'FontSize',7,'HorizontalAlignment','center');

text(ax, 0, fh+14,'Ti top wall','Color',[0.55 0.92 0.68],'FontSize',7,...
     'HorizontalAlignment','center');

%% Create patches
patch_p = gobjects(ne_p,1);
for e=1:ne_p
  en=elem_p(e,:);
  patch_p(e) = patch(ax, Xp(en)*1000, Yp(en)*1000, [0.4 0.2 0.1], ...
      'EdgeColor',[0.07 0.13 0.22],'LineWidth',0.2);
end

patch_f = gobjects(ne_f,1);
for e=1:ne_f
  en=elem_f(e,:);
  patch_f(e) = patch(ax, Xf(en)*1000, Yf(en)*1000, [0.3 0.4 0.3], ...
      'EdgeColor',[0.05 0.14 0.10],'LineWidth',0.2);
end

cb = colorbar(ax,'eastoutside');
cb.Color=[0.65 0.75 0.90];
cb.Label.String='Von Mises stress (normalised)';
cb.Label.Color=[0.65 0.75 0.90];

%% HUD displays
htxt_phase = text(ax,0,0,'Impact simulation running...','Color',[0.88 0.95 1.0],...
    'FontSize',10,'FontWeight','bold','Units','normalized','Position',[0.02 0.99]);
htxt_stats = text(ax,0,0,'','Color',[0.95 0.82 0.30],'FontSize',8.5,...
    'FontName','Courier New','Units','normalized','Position',[0.02 0.91]);
htxt_err = text(ax,0,0,'','Color',[0.28 0.95 0.56],'FontSize',9.5,...
    'FontWeight','bold','Units','normalized','Position',[0.02 0.82]);

%% Live plots
h_def_p  = plot(ax2,NaN,NaN,'-','Color',[0.85 0.65 0.35],'LineWidth',2.2);
h_def_f  = plot(ax4,NaN,NaN,'-','Color',[0.38 0.95 0.55],'LineWidth',2.2);
h_error  = plot(ax3,NaN,NaN,'-','Color',[0.95 0.75 0.20],'LineWidth',2.2);
h_dot2   = plot(ax2,NaN,NaN,'o','Color','w','MarkerSize',5);
h_dot4   = plot(ax4,NaN,NaN,'o','Color','w','MarkerSize',5);
h_dot3   = plot(ax3,NaN,NaN,'o','Color','w','MarkerSize',5);

plot_every = max(1, floor(Nsteps/600));
cmap = colormap(ax);
ncolors = size(cmap,1);

%% ===== MAIN TIME LOOP =====
fprintf('Running Copper panel impact simulation...\n\n');

for step = 1:Nsteps
  t = step*dt;

  %% Panel (COPPER) internal forces
  fi_p = zeros(nn_p,2);
  for e=1:ne_p
    en=elem_p(e,:);
    ue=reshape([up(en,1) up(en,2)]',8,1);
    eps=Bp*ue; sig=Dm_cu*eps; fe=Bp'*sig*Ve_p;
    for k=1:4
      fi_p(en(k),1)=fi_p(en(k),1)+fe(2*k-1);
      fi_p(en(k),2)=fi_p(en(k),2)+fe(2*k);
    end
    s1=sig(1); s2=sig(2); s3=sig(3);
    svm_p(e)=sqrt(s1^2-s1*s2+s2^2+3*s3^2);
  end

  %% Frame internal forces
  fi_f = zeros(nn_f,2);
  for e=1:ne_f
    en=elem_f(e,:);
    ue=reshape([uf(en,1) uf(en,2)]',8,1);
```

```
  eps=Bf*ue; sig=Dm_ti*eps; fe=Bf'*sig*Ve_f;
  for k=1:4
    fi_f(en(k),1)=fi_f(en(k),1)+fe(2*k-1);
    fi_f(en(k),2)=fi_f(en(k),2)+fe(2*k);
  end
  s1=sig(1); s2=sig(2); s3=sig(3);
  svm_f(e)=sqrt(s1^2-s1*s2+s2^2+3*s3^2);
end

%% Contact forces
fc_p = zeros(nn_p,2);
if ~done1
  nx_n=Xp+up(:,1); ny_n=Yp+up(:,2);
  d2=(nx_n-dX1).^2+(ny_n-dY1).^2;
  in_c=d2<R_d^2 & d2>1e-20;
  if any(in_c)
    dv=sqrt(d2(in_c)); fn=k_pen_cu*(R_d-dv);
    nhx=(nx_n(in_c)-dX1)./dv; nhy=(ny_n(in_c)-dY1)./dv;
    fc_p(in_c,1)=fn.*nhx; fc_p(in_c,2)=fn.*nhy;
    dVx1=dVx1-sum(fn.*nhx)/m_d*dt;
    dVy1=dVy1-sum(fn.*nhy)/m_d*dt;
    if ~contact1
      wave_r1=1e-4; imp_x1=dX1; imp_y1=dY1;
      contact1=true;
      fprintf('  IMPACT #1 (COPPER Panel) at t=%.2f μs\n', t*1e6);
    end
  end
end

fc_f = zeros(nn_f,2);
if ~done2
  nx_f2=Xf+uf(:,1); ny_f2=Yf+uf(:,2);
  d2f=(nx_f2-dX2).^2+(ny_f2-dY2).^2;
  in_f=d2f<R_d^2 & d2f>1e-20;
  if any(in_f)
    dvf=sqrt(d2f(in_f)); fnf=k_pen_f*(R_d-dvf);
    nhxf=(nx_f2(in_f)-dX2)./dvf; nhyf=(ny_f2(in_f)-dY2)./dvf;
    fc_f(in_f,1)=fnf.*nhxf; fc_f(in_f,2)=fnf.*nhyf;
    dVx2=dVx2-sum(fnf.*nhxf)/m_d*dt;
    dVy2=dVy2-sum(fnf.*nhyf)/m_d*dt;
    if ~contact2
      wave_r2=1e-4; imp_x2=dX2; imp_y2=dY2;
      contact2=true;
      fprintf('  IMPACT #2 (Ti Frame) at t=%.2f μs\n', t*1e6);
    end
  end
end

%% Velocity-Verlet updates
acc_p=(-fi_p+fc_p)./[Mp Mp];
acc_p(isnan(acc_p))=0;
acc_p(bc_p,:)=0;
vp=vp+acc_p*dt; up=up+vp*dt; up(bc_p,:)=0; vp(bc_p,:)=0;

acc_f=(-fi_f+fc_f)./[Mf Mf];
acc_f(isnan(acc_f))=0;
acc_f(bc_f,:)=0;
vf=vf+acc_f*dt; uf=uf+vf*dt; uf(bc_f,:)=0; vf(bc_f,:)=0;

%% Debris motion
dX1=dX1+dVx1*dt; dY1=dY1+dVy1*dt;
if dY1<py0-0.05, done1=true; end
dX2=dX2+dVx2*dt; dY2=dY2+dVy2*dt;
if dY2<fy0-0.03, done2=true; end

%% Wave propagation
if wave_r1>0, wave_r1=wave_r1+cw_cu*dt; end
if wave_r2>0, wave_r2=wave_r2+cw_ti*dt; end

%% Record history
md_p=max(sqrt(up(:,1).^2+up(:,2).^2))*1000;
md_f=max(sqrt(uf(:,1).^2+uf(:,2).^2))*1000;
if md_p>max_def_p, max_def_p=md_p; end
if md_f>max_def_f, max_def_f=md_f; end
def_hist_p(step)=md_p; def_hist_f(step)=md_f;
t_hist(step)=t;

%% Animation
```

```
  if mod(step,plot_every)==0 || step==Nsteps
    % Update colors
    sv_max_p=max(svm_p);
    if sv_max_p<1e-12, sv_max_p=1; end
    for e=1:ne_p
      en=elem_p(e,:);
      xv=(Xp(en)+up(en,1))*1000; yv=(Yp(en)+up(en,2))*1000;
      stress_norm=max(0,min(1,svm_p(e)/sv_max_p));
      color_idx=max(1,min(ncolors,round(stress_norm*(ncolors-1))+1));
      set(patch_p(e),'XData',xv,'YData',yv,'FaceColor',cmap(color_idx,:));
    end

    sv_max_f=max(svm_f);
    if sv_max_f<1e-12, sv_max_f=1; end
    for e=1:ne_f
      en=elem_f(e,:);
      xv=(Xf(en)+uf(en,1))*1000; yv=(Yf(en)+uf(en,2))*1000;
      stress_norm=max(0,min(1,svm_f(e)/sv_max_f));
      color_idx=max(1,min(ncolors,round(stress_norm*(ncolors-1))+1));
      set(patch_f(e),'XData',xv,'YData',yv,'FaceColor',cmap(color_idx,:));
    end

    % Display results
    set(htxt_stats,'String',sprintf(...
        't = %6.1f μs  |  step %5d\n  COPPER Panel: %7.3f mm  |  Ti Frame: %6.3f mm',...
        t*1e6, step, max_def_p, max_def_f));

    set(htxt_err,'String',sprintf('Maximum deflection: %.2f mm', max_def_p), ...
        'Color', [0.28 0.95 0.56]);

    % Update plots
    idx_v=1:step;
    set(h_def_p,'XData',t_hist(idx_v)*1e6,'YData',def_hist_p(idx_v));
    set(h_def_f,'XData',t_hist(idx_v)*1e6,'YData',def_hist_f(idx_v));
    set(h_dot2,'XData',t*1e6,'YData',md_p);
    set(h_dot4,'XData',t*1e6,'YData',md_f);

    drawnow limitrate;
  end
end

%% ===== FINAL RESULTS =====
fprintf('\n============================================\n');
fprintf('  SIMULATION RESULTS\n');
fprintf('============================================\n');
fprintf('  COPPER Panel:\n');
fprintf('    Max deflection  : %.3f mm\n', max_def_p);
fprintf('\n  TITANIUM Frame:\n');
fprintf('    Max deflection  : %.3f mm\n', max_def_f);
%fprintf('============================================\n");
fprintf('\n✓ Simulation complete!\n');
```